AIM: Study of Different Types of Network Cables

OBJECTIVE

To study and understand the different types of network cables used in computer networking, their categories, data transmission speeds, advantages, disadvantages, and applications.

INTRODUCTION

Network cables are physical communication media used to connect computers, network devices, servers, and other networking equipment. They are used to transmit data from one device to another. Different types of cables are selected depending on factors such as transmission speed, distance, cost, bandwidth, and environmental conditions.

The major types of network cables are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable

1. UNHSHIELDED TWISTED PAIR (UTP) CABLE

UTP cable consists of pairs of insulated copper wires twisted together to reduce electromagnetic interference. It is one of the most commonly used cables in LAN networks.

Categories and Applications

Category	Maximum Data Transmission	Application / Use
Category 3 (Cat 3)	Up to 10 Mbps	10Base-T Ethernet and telephone networks

Category 5 (Cat 5)	Up to 100 Mbps	Fast Ethernet
Category 5e (Cat 5e)	Up to 1 Gbps	Gigabit Ethernet
Category 6 (Cat 6)	Up to 10 Gbps	Gigabit Ethernet and high-speed networks
Category 6a (Cat 6a)	Up to 10 Gbps	10G Ethernet

Advantages

- Low cost and economical.
- Easy to install and maintain.
- Lightweight and flexible.
- Smaller overall diameter.
- Widely available.

Disadvantages

- More susceptible to electromagnetic interference (EMI).
- Signal quality may decrease over long distances.
- Provides less protection against external noise compared to shielded cables.

Applications

- Local Area Networks (LAN).
- Ethernet connections.
- Computer-to-switch connections.
- Telephone and data communication networks.

2. SHIELDED TWISTED PAIR (STP) CABLE

STP cable is a twisted-pair cable that has an additional metallic shield around the wires. The shielding provides better protection against electromagnetic interference and electrical noise.

Categories and Applications

Category	Maximum Data Transmission	Application / Use
Cat 6	Up to 10 Gbps	Gigabit Ethernet
Cat 6a	Up to 10 Gbps	10G Ethernet
Cat 7	Up to 10 Gbps	High-speed Ethernet and data centres

Advantages

- Better protection from electromagnetic interference.
- Less affected by electrical noise.
- Provides reliable data transmission.
- Suitable for environments with high electrical interference.
- Supports high-speed networking.

Disadvantages

- More expensive than UTP.
- Installation is comparatively difficult.
- Thicker and less flexible.
- Requires proper grounding for effective shielding.

Applications

- Data centres.
- Industrial networks.
- High-speed Ethernet networks.
- Areas with high electromagnetic interference.

3. COAXIAL CABLE

Coaxial cable consists of a central copper conductor, an insulating layer, a metallic shield, and an outer protective jacket. It provides good protection against interference and has been widely used for television and communication systems.

Types of Coaxial Cable

Type	Typical Application
RG-6	Cable television and high-speed internet
RG-59	CCTV and television systems
RG-11	Long-distance cable television and broadband connections

Maximum Data Transmission

Coaxial cable can support data rates ranging from approximately **10 Mbps to 100 Mbps**, depending on the cable type and network technology.

Advantages

- Good resistance to electromagnetic interference.
- Reliable signal transmission.
- Good bandwidth.
- Low signal loss.
- Durable and versatile.

Disadvantages

- Bulky compared with twisted-pair cables.
- More difficult to install.
- Higher cost in some applications.
- Limited flexibility.

Applications

- Cable television networks.
- Broadband internet connections.
- CCTV systems.
- Radio and communication systems.

4. FIBRE OPTIC CABLE

Fibre optic cable transmits data in the form of light signals through thin strands of glass or plastic fibre. It provides very high bandwidth and is suitable for long-distance and high-speed communication.

Types of Fibre Optic Cable

Type	Description	Application
Single-Mode Fibre	Uses a single light path and supports long-distance communication	Long-distance networks, telecom and WAN
Multi-Mode Fibre	Uses multiple light paths and is suitable for shorter distances	LANs, data centres and campus networks

Data Transmission

Fibre optic cables can support **very high data rates, including 100 Gbps and beyond**, depending on the equipment and fibre technology used.

Advantages

- Very high transmission speed.
- High bandwidth.
- Supports long-distance communication.
- Immune to electromagnetic interference.
- Provides better security than copper cables.
- Low signal attenuation.
- Lightweight compared with many copper cable systems.

Disadvantages

- Higher installation cost.
- Requires skilled technicians for installation and maintenance.
- Fibre can be more delicate than copper cables.
- Special equipment is required for splicing and testing.

Applications

- Internet backbone networks.

- Data centres.
- Telecommunication networks.
- Long-distance communication.
- High-speed LAN and WAN networks.

COMPARISON OF DIFFERENT NETWORK CABLES

Cable Type	Medium	Speed	Interference Protection	Cost	Common Applications
UTP	Copper	Up to 10 Gbps	Low to Medium	Low	LAN, Ethernet
STP	Shielded Copper	Up to 10 Gbps	High	Medium	Data centres, industrial networks
Coaxial	Copper	Up to 100 Mbps*	High	Medium	TV, CCTV, broadband
Fibre Optic	Glass/Plastic Fibre	100 Gbps and beyond*	Very High	High	Internet, telecom, data centres

*Actual speed depends on the specific cable, equipment, standard, and network configuration.